1. Title:

Design and Finite Element Optimization of a Bio-Inspired Force-Sensing Brain Biopsy Needle for Minimizing Tissue Damage and Improving Targeting Accuracy.

2. Problem Statement:

Conventional rigid bevel-tip brain biopsy needles can damage nearby brain tissue during insertion. This may cause the brain to shift slightly, making it harder to reach the exact target identified in MRI or CT scans.Increasing the risk of bleeding, tissue injury, and collecting samples from the wrong location.

3. Objectives

- Design a conventional brain biopsy needle and a novel bio-inspired biopsy needle using SolidWorks.
- Develop different bio-inspired tip geometries to reduce insertion force and tissue trauma.
- Perform finite element analysis (ANSYS) to evaluate:
- Stress distribution
- Deformation
- Buckling resistance
- Contact pressure with brain tissue
- Compare the mechanical performance of the proposed design with conventional biopsy needles.

4. Literature Survey:

1. MRI-Compatible Fiber-Optic Shape-Sensing Biopsy Needle

What researchers did:

- They developed a biopsy needle that can be safely used inside an MRI scanner.
- Tiny optical fibers (fiber Bragg gratings) are placed inside the needle.
- These fibers measure how much the needle bends during insertion.
- The system estimates the needle's shape in real time, helping surgeons know its exact position.

Limitation:

- It mainly tracks needle bending.
- It does not reduce tissue damage or improve the needle tip design.

2. Mosquito-Inspired Needle

What researchers did:

- Researchers studied how mosquitoes insert their proboscis with very little pain.
- They proposed features such as:
  - Serrated (saw-like) needle edges

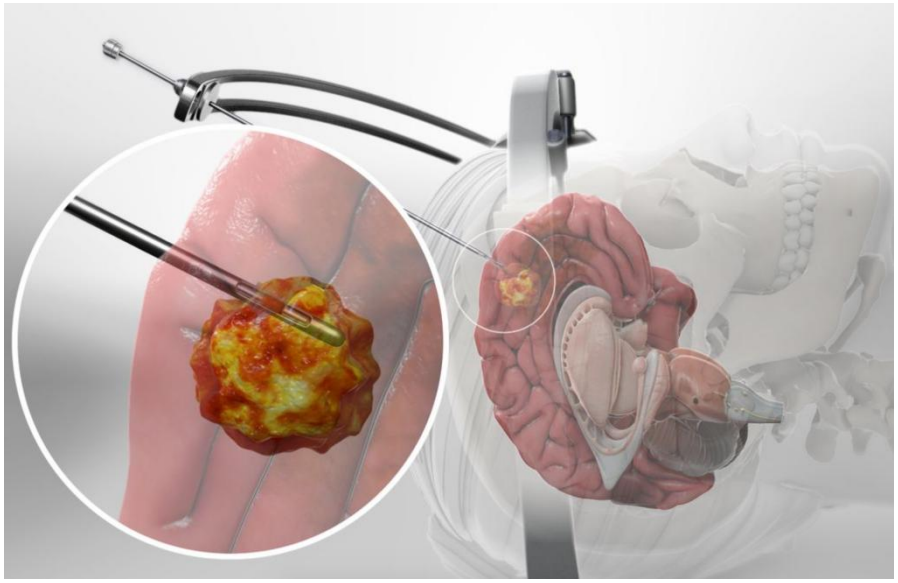
- Small vibrations during insertion

- These features reduce insertion force and improve accuracy.

Limitation:

- Most studies are still experimental.
- Very few have applied these ideas specifically to brain biopsy needles.
- Comparative finite element analysis (FEA) of these designs is limited.

Prior art and previous research:





Research Gap:

- Conventional brain biopsy needles use standard tip geometries, which can cause high insertion force, tissue deformation, and unnecessary damage to healthy brain tissue.
- Limited research integrates bio-inspired tip geometry with mechanical optimization, leaving opportunities to improve targeting accuracy and reduce needle deflection.

- Few studies perform a comprehensive finite element comparison of different needle designs and biomaterials to optimize stress distribution, stiffness, and tissue interaction in a single framework.

5. Proposed Solution

Concept Generation

Two bio-inspired needle designs were considered:

Option 1: Mosquito-Inspired Needle

- Inspired by the way a mosquito inserts its proboscis with very little pain.
- The needle uses tiny vibrations and a serrated (saw-like) tip to reduce insertion force.
- It mainly helps reduce tissue damage but cannot improve steering inside the brain.

Option 2: Wasp-Inspired Needle (Selected Design)

- Inspired by the egg-laying organ (ovipositor) of a wasp.
- The needle has 2–3 thin sliding segments that move back and forth during insertion.
- This allows the needle to enter brain tissue with less force and better control.
- It also improves targeting accuracy by reducing needle bending. Selected Concept: The wasp-inspired segmented needle was chosen because it can both reduce tissue damage and improve targeting accuracy.

Design Methodology

- Design a conventional bevel-tip biopsy needle to use as a reference model.
- Design a bio-inspired segmented biopsy needle with 2–3 sliding segments based on the wasp ovipositor.
- Add a strain sensor near the base of the needle to measure the forces acting on the needle during insertion.
- Create both models in SolidWorks and import them into ANSYS.
- Simulate insertion into a brain tissue model and compare:
    - Stress
    - Deformation
    - Tissue strain
    - Needle bending
    - Insertion force
- Optimize design parameters such as:
    - Segment overlap
    - Tip angle
    - Sliding movement
- Select the design that provides minimum tissue damage and maximum targeting accuracy.

Novelty

- Bio-inspired needle tip geometry.
- Mechanical optimization through finite element analysis.

- Optimizing the needle design to achieve both lower tissue damage and better targeting accuracy.
- Comparative evaluation against existing biopsy needle designs.

6. Implant Design:

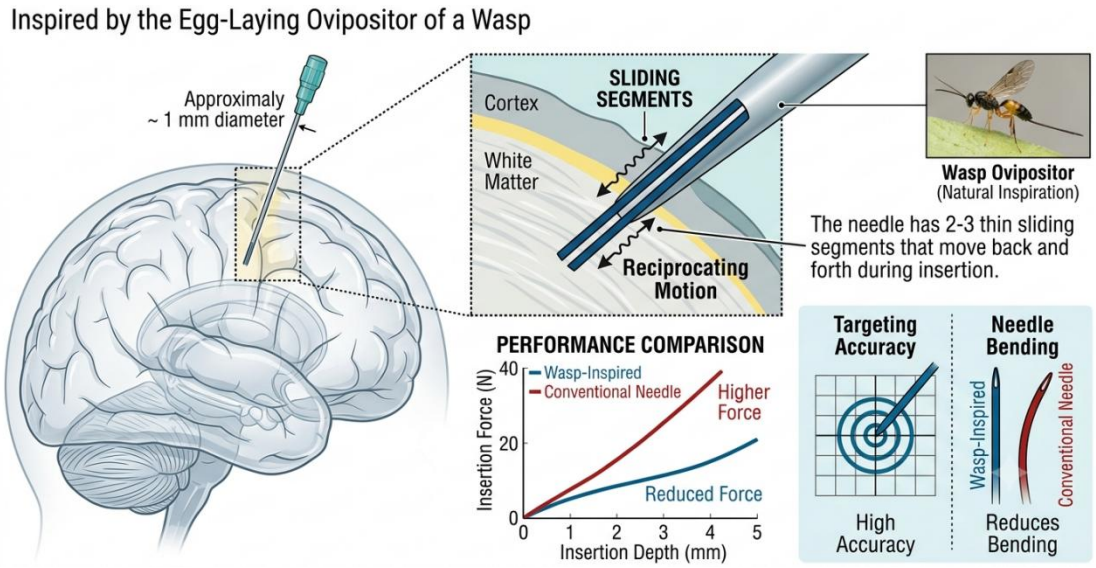


Fig: proposed design (Gemini Image)

7. Comparison with Existing Designs

Design	Strength	Biocompatibility	Ergonomics	Manufacturability	Cost
Conventional rigid bevel-tip needle	9	7	5	9	7
Programmable Bevel-Tip Needle	5	7	7	3	3
FBG optical shape-sensing needle	7	7	5	3	1
Concentric-tube needle w/ resistivity + force sensing	5	5	5	3	3
Proposed design	7	7	7	-	5